



variables (non-basic variables) at zero level and then solving for them. This solution is called the initial basic feasible solution.

To move from one basic solution to another row operations using Gauss-Jordan elimination method is employed. Successive iteration yields a better solution. This continues until the optimal solution is reached (solution which cannot be improved further). The process of iteration is guided by two conditions:

OPTIMALITY CONDITION:- This condition determines when the iteration should stop. It selects an entering variable from among the non-basic variables as one that has the most negative coefficient for a maximization problem (or most positive coefficient for minimization problem) of the objective function. Select any one where more than one qualifies. The tableau becomes optimal when all the coefficient of the objective row have been transformed to either positive or zeros (for maximization) or negatives or zeros (in case of minimization).

FEASIBILITY CONDITION: This selects a leaving variable from among the basic variables as one that first reach the zero level (becoming non basic variable). This is to give room for the incoming variable. This is done by taking the ratio of the right hand side (R.H.S.) to the elements in the corresponding entering column. The minimum is then identified and the row associated to



Candidate's Number

Question

Write on both sides of the paper

Date: 04/07/2020
Page: 7

It leaves. If more than one minima exist, select any one.

N.B: Ratios are not taken for Zero or negative R.H.S.

The Steps of the Simplex method are:

Step 0. Determine a Starting basic feasible solution.

Step 1: Select an entering variable using the optimality condition. Stop if there is no entering variable; last solution is optimal.

Step 2: Select a leaving variable using the feasibility condition.

Step 3: Determine the new basic solution by using the appropriate Gauss-Jordan computations. Go to Step 1.

Example:

Using Simplex method:

$$\begin{aligned} \text{Maximize } Z &= 5x_1 + 4x_2 \\ \text{Subject to } &6x_1 + 4x_2 \leq 24 \\ &x_1 + 2x_2 \leq 6 \\ &-x_1 + x_2 \leq 1 \\ &x_2 \leq 2 \\ &x_1, x_2 \geq 0. \end{aligned}$$

Solution:

Standard form:

$$\text{Maximize } Z = 5x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 + 0s_4.$$

st:

$$\begin{aligned} 6x_1 + 4x_2 + s_1 &= 24 \\ x_1 + 2x_2 + s_2 &= 6 \\ -x_1 + x_2 + s_3 &= 1 \\ x_2 + s_4 &= 2 \end{aligned}$$

$$x_1, x_2, s_1, s_2, s_3, s_4 \geq 0$$



Candidate's Number

Question

Write on both sides of the paper

The variables S_1, S_2, S_3 and S_4 are the Slacks associated with the respective constraints.

Next

we express the objective equation as

$$Z - 5x_1 - 4x_2 = 0$$

In this manner, the Starting Simplex tableau can be ~~rep~~ represented as follows:

nonbasic variables (x_1, x_2)

Basic variables (S_1, S_2, S_3, S_4).

Given that the nonbasic variable $(x_1, x_2) = (0, 0)$

then $Z = 0$

$$S_1 = 24$$

$$S_2 = 6$$

$$S_3 = 1$$

$$S_4 = 2$$

	x_1	x_2	S_1	S_2	S_3	S_4	RHS.	Ratio
Z	-5	-4	0	0	0	0	0	
S_1	6	4	1	0	0	0	24	$\frac{24}{6} = 4$
S_2	1	2	0	1	0	0	6	$\frac{6}{1} = 6$
S_3	-1	1	0	0	1	0	1	$\frac{1}{-1} = -1$
S_4	0	1	0	0	0	1	2	$\frac{2}{0} = \infty$

* The entering variable is x_1 because it has the most negative coefficient in the (max) obj. fn.

~~na~~ If all the obj. fn. coefficient happens to be ≥ 0 , no further improvement in Z is possible signifying that the optimal has been reached.

* The leaving variable is the ratio of the R.H.S. of the equation to the corresponding



Candidate's Number

Question

Write on both sides of the paper

021718
Do not write
in either
margin

constraint coefficients under the entering variable x_i .

The minimum nonnegative ratio corresponds to basic S_i , signifying that S_i is the leaving variable.

The pivot column and pivot row ^{is associated with} ~~with~~ the entering and the leaving variables respectively. The intersection of the pivot column and the pivot row is called the pivot element.

	x_1	x_2	S_1	S_2	S_3	S_4	RHS.
Basic Z	-5	-4	0	0	0	0	0
S_1	(6)	4	1	0	0	0	24
S_2	1	2	0	1	0	0	6
S_3	-1	1	0	0	1	0	1
Ratio S_4	0	1	0	0	0	1	2

Pivot element is 6

Pivot column is x_1

Pivot row is S_1

The Gauss-Jordan Computations needed to produce the new basic solution include two types

(1) Pivot row

New pivot row = Current pivot row $\div$ pivot element

(2) All other rows including Z

New Row = (Current row) - its pivot column coefficient $\times$ (New pivot row)

These computation are applied to the preceding tableau in the following manner



Candidate's Number: _____

Question _____

Write on both sides of the paper

- (a) New pivot S_1 -row = Current $S_1 \div 6$
- (b) New Z-row = Current Z-row - $(-5) \times$ New pivot row
- (c) New S_2 -row = Current S_2 -row - $(1) \times$ New pivot row
- (d) New S_3 -row = Current S_3 -row - $(-1) \times$ New pivot row
- (e) New S_4 -row = Current S_4 -row - $(0) \times$ New pivot row.

The new tableau corresponding to the new basic solution (x_1, s_2, s_3, s_4) and nonbasic variable (s_1, x_2)

	x_1	x_2	s_1	s_2	s_3	s_4	RHS.
Z Basic	0	$-\frac{2}{3}$	$\frac{5}{6}$	0	0	0	20
x_1	1	$\frac{2}{3}$	$\frac{1}{6}$	0	0	0	4
s_2	0	$\frac{4}{3}$	$-\frac{1}{6}$	1	0	0	2
s_3	0	$\frac{5}{3}$	$\frac{1}{6}$	0	1	0	5
s_4	0	1	0	0	0	1	2

Using the same procedures,
 x_2 will be the entering variable
 and s_2 is the leaving variable.

The computations produce

	x_1	x_2	s_1	s_2	s_3	s_4	RHS.
Z Basic	0	0	$\frac{3}{4}$	$\frac{1}{2}$	0	0	21
x_1	1	0	$\frac{1}{4}$	$-\frac{1}{2}$	0	0	3
x_2	0	1	$-\frac{1}{8}$	$\frac{3}{4}$	0	0	$\frac{3}{2}$
s_3	0	0	$\frac{3}{8}$	$-\frac{5}{4}$	1	0	$\frac{5}{2}$
s_4	0	0	$\frac{1}{8}$	$-\frac{3}{4}$	0	1	$\frac{1}{2}$

Because none of the Z-row ~~off~~ coefficients associated with the nonbasic variable s_1 and s_2 are negative. The table is optimal



Candidate's Number.....

Question.....

Write on both sides of the paper

2017
Do not write
in either
margin

The optimum solution can be read from the simplex tableau in the following manner.

Decision variable	Optimum value
x_1	3
x_2	$3/2$
Z	21

You can verify the values $s_1 = s_2 = 0$, $s_3 = 5/2$, $s_4 = 1/2$ are consistent with the given values of x_1 and x_2 .